

ABERDARE TECHNICAL COLLEGE

**NETWORK SETUP
ELEMENT 2: CONNECT NETWORK DEVICES**

ISCED UNIT CODE: 0612 351 03A
PRACTICAL 2

Candidate name: _____ **Admission no.:** _____

Class/Course: _____ **Date:** _____

Duration: 2 hours

Instructions

- *Attempt all tasks in the order given.*
- *Work individually and use only the equipment provided.*
- *Record your findings on the answer sheet at the end.*
- *Call the examiner to observe and sign off at each checkpoint.*
- *You have been provided with the following materials, a router, 3 pcs, 1 switch, terminated and Ethernet cables.*
- *Ensure you have all the materials before commencing*

Equipment provided

3 PCs, 1 switch, 1 router, 4 Ethernet patch cables, router default login sheet.

Target addressing table

Device	IP address	Subnet mask	Default gateway
Router (new)	192.168.50.1	255.255.255.0	n/a
PC1	192.168.50.11	255.255.255.0	192.168.50.1
PC2	192.168.50.12	255.255.255.0	192.168.50.1
PC3	192.168.50.13	255.255.255.0	192.168.50.1

Task 1: Set up the network hardware

1. Connect the router's LAN port to the switch.
2. Connect PC1, PC2 and PC3 to separate ports on the switch.
3. Power on the router, switch and all three PCs.
4. Check that the link lights are on for every connected port.
5. Record the number of cables used and the switch ports occupied.

Task 2: Access the router

1. On PC1, open Command Prompt and run ipconfig.
2. Identify and record the default gateway.
3. Enter the gateway address in a web browser.
4. Log in to the router using the credentials provided.
5. Record the router's default IP address.

Task 3: Change the router's IP address

1. Open the router's LAN settings.
2. Change the router's IP address to 192.168.50.1 with subnet mask 255.255.255.0.
3. Set the DHCP range to 192.168.50.100 to 192.168.50.150 (if DHCP is enabled).
4. Save the settings and wait for the router to restart.
5. Reconnect and open 192.168.50.1 in the browser to confirm the change.
6. Record the old IP address and the new IP address.

Task 4: Configure IP addresses on the PCs

1. On each PC, open the IPv4 properties of the Ethernet adapter.
2. Assign the IP address, subnet mask and default gateway shown in the addressing table.
3. Set the DNS server to 8.8.8.8.
4. Run ipconfig on each PC to verify the settings.
5. Record the configuration of each PC.

Task 5: Test connectivity

1. From PC1, ping the router (192.168.50.1).
2. From PC1, ping PC2 and PC3.
3. From PC2, ping PC3.
4. Record the result of each test.
5. If any test fails, troubleshoot (cables, IP settings, firewall) and record the fix.

Task 6: Prepare the PCs for file sharing

1. On each PC, open Advanced sharing settings.
2. Turn on network discovery and file and printer sharing.
3. Set the network profile to Private.

Task 7: Share a folder on PC1

1. Create a folder named **ExamShare** on the PC1 Desktop.
2. Inside it, create a text file **candidate.txt** containing your name and admission number.
3. Share the folder using the share name **ExamShare**.
4. Give **Everyone** Read permission.

Task 8: Access shared files across the network

1. From PC2, open \\192.168.50.11, open **ExamShare**, and copy candidate.txt to the PC2 Desktop.
2. Repeat the same steps from PC3.
3. On PC2, create a folder **FromPC2** containing **hello.txt**, and share it with Read/Write permission.
4. From PC3, open \\192.168.50.12 and copy hello.txt to the PC3 Desktop.

The end.

The trainer is introducing oral questions shortly